Baiting Luo

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Education

Vanderbilt University, Tennessee — Aug 2021 – Aug 2026
Ph.D. in Computer Science
Advisor: Prof. Abhishek Dubey

Northwestern University, Illinois — Sept 2019 – June 2021
M.S. in Computer Engineering
Thesis: Sybil Attack Detection in VANET (Northwestern Best CE Thesis Award)
Advisor: Prof. Qi Zhu

Rensselaer Polytechnic Institute (RPI), New York — Sept 2015 – May 2019
B.S. in Computer Engineering
Dual Degree: Computer Science

Work Experience

Research Topics: Online/Offline Reinforcement Learning, Decision Making under Uncertainty, Planning, Machine Learning, Heuristic Search, Autonomous Cyber-Physical Systems

Research Intern, **Mentor:** Dr. Arun Ramamurthy — May 2025 – Aug 2025
Siemens Corporation, Technology

- **Sample-Efficient World Action Model:**
  - Developed ARGO, a world action model agent that learns strong control policies from fewer real environment interactions by planning in a learned model, and uses model uncertainty asymmetrically: an ensemble-based optimistic planner explores the real environment, while replay-grounded, pessimistic actor updates prevent model exploitation from biasing the policy. Improved aggregate visual DeepMind Control Suite performance by 8.9% over the strongest recurrent world model baseline and 13.3% over the Transformer world model backbone at 1M environment steps. (*Under Review*)

Ph.D. Student and Research Assistant, **Advisor:** Prof. Abhishek Dubey — Aug 2021 – Aug 2026
Institute for Software Integrated Systems, Vanderbilt University

- **Planning with Latent World Models for Adaptive High-Dimensional Control:**
  - Developed Latent Macro Action Planner (L-MAP) for offline reinforcement learning, leveraging a state-conditioned VQ-VAE to learn temporally extended macro-actions for autoregressive sequence modeling, combined with Monte Carlo Tree Search (MCTS) for stochastic planning; achieved state-of-the-art performance on high-dimensional continuous control tasks while enabling low-latency decision-making in stochastic environments. (*ICLR 2025 Spotlight*)[paper][code]
  - Developed an in-context planning agent that discretizes observation–macro-action sequences (RQ-VAE) and plans with MCTS in latent space using a GPT-style temporal Transformer prior. Trained offline, adapts online under stochastic, partially observable dynamics; matches or outperforms strong baselines on MuJoCo/Adroit with low decision latency. (*Under Review*)[paper][code]
- **Decision Making in Non-Stationary Environment:**
  - Proposed Adaptive MCTS for safe exploration and online adaptation to changing dynamics in model-based reinforcement learning tasks. (*AAMAS 2024*)[paper][code].
  - Created NS-Gym toolkit for standardized evaluation of online decision-making algorithms in dynamically changing environments. (*NeurIPS 2025 D&B Track*)[paper][code].
- **Runtime Safety Assurance of Autonomous Vehicles:**
  - Proposed Dynamic Simplex framework improving performance without compromising safety in autonomous systems through planning with multiple generative models in dynamic environments (*ICCPS 2023*)[paper][code].
  - Developed advanced sampling techniques for high-risk scenario generation in AV testing (*ICAA 2022*)[paper]

[code].

- Created an automated testing framework for adversarial conditions in AV simulations (*ITSC 2022*) [paper] [code].

Research Assistant, **Advisor:** Prof. Qi Zhu Mar 2020 – July 2021
Design Automation of Intelligent Systems Lab, Northwestern University

- **Securing Connected and Autonomous Vehicles:**
  - Developed hybrid GCN-RNN model to detect Sybil attacks in connected vehicle networks (*IV 2021*) [paper].
  - Created dual cyber-physical blockchain framework for efficient security in large-scale vehicular networks (*IV 2021*) [paper].

Publications

** indicates equal contribution*

Baiting Luo, Ava Pettet, Aron Laszka, Abhishek Dubey, Ayan Mukhopadhyay, "Scalable Decision-Making in Stochastic Environments through Learned Temporal Abstraction", *13th International Conference on Learning Representations (**ICLR 2025 Spotlight**)*. (**Top 5.1%**) [code]

Baiting Luo, Nathaniel S Keplinger, Yunuo Zhang, Arun Ramamurthy, Hyunjee Jin, Ayan Mukhopadhyay, Abhishek Dubey, "ARGO: Asymmetric Replay Grounding with Optimistic Planning". (Under Review)

Baiting Luo, Yunuo Zhang, Nathaniel S Keplinger, Samir Gupta, Abhishek Dubey, Ayan Mukhopadhyay, "In-Context Planning with Latent Temporal Abstractions". (Under Review)

Baiting Luo, Yunuo Zhang, Abhishek Dubey, Ayan Mukhopadhyay, "Act as You Learn: Adaptive Decision-Making in Non-Stationary Markov Decision Processes", *23rd International Conference on Autonomous Agents and MultiAgent Systems (AAMAS 2024)*. (Acceptance rate: 25%) (**Vanderbilt C.F. Chen Best Paper Runner-Up Award**) [code]

Baiting Luo, Shreyas Ramakrishna, Ava Pettet, Christopher Kuhn, Gabor Karsai, Ayan Mukhopadhyay, "Dynamic Simplex: Balancing Safety and Performance in Autonomous Cyber Physical Systems", *14th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS 2023)*. (Acceptance rate: 25.6%) [code]

Baiting Luo*, Shreyas Ramakrishna*, Christopher Kuhn, Gabor Karsai, Abhishek Dubey, "ANTI-CARLA: An Adversarial Testing Framework for Autonomous Vehicles in CARLA", *IEEE 25th International Conference on Intelligent Transportation Systems (ITSC 2022)*. [code]

Baiting Luo, Xiangguo Liu, Qi Zhu, "Credibility Enhanced Temporal Graph Convolutional Network Based Sybil Attack Detection On Edge Computing Servers", accepted by *32nd IEEE Intelligent Vehicles Symposium (IV 2021)*. (Acceptance rate: 49.3%)

Baiting Luo, "Adaptive Decision-Making in Non-Stationary Markov Decision Processes", Doctoral Consortium in the *23rd International Conference on Autonomous Agents and MultiAgent Systems (AAMAS 2024)*.

Yunuo Zhang, **Baiting Luo**, Ayan Mukhopadhyay, Gabor Karsai, Abhishek Dubey, "ESCORT: Efficient Stein-variational and Sliced Consistency-Optimized Temporal Belief Representation for POMDPs", *39th Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)*. (Acceptance rate: 24.52%)

Nathaniel S. Keplinger, **Baiting Luo**, Iliyas Bektas, Yunuo Zhang, Kyle Hollins Wray, Aron Laszka, Abhishek Dubey, Ayan Mukhopadhyay, "NS-Gym: Open-Source Simulation Environments and Benchmarks for Non-Stationary Markov Decision Processes", *39th Annual Conference on Neural Information Processing Systems (**NeurIPS 2025 Datasets & Benchmarks Track**)*. (Acceptance rate: 24.91%) [code]

Yunuo Zhang, **Baiting Luo**, Ayan Mukhopadhyay, Abhishek Dubey, "Observation Adaptation via Annealed Importance Resampling for Partially Observable Markov Decision Processes", *International Conference on Automated Planning and Scheduling (ICAPS 2025)*. (Acceptance rate: 22.8%)

Yunuo Zhang, **Baiting Luo**, Ayan Mukhopadhyay, Daniel Stojcsics, Daniel Elenius, Anirban Roy, Susmit Jha, Miklos Maroti, Xenofon Koutsoukos, Gabor Karsai, Abhishek Dubey, "Shrinking POMCP: A Framework for Real-Time UAV Search and Rescue", *IEEE International Conference on Assured Autonomy (ICAA 2024)*.

Ava Pettet, Yunuo Zhang, **Baiting Luo**, Kyle Wray, Hendrik Baier, Aron Laszka, Abhishek Dubey, Ayan Mukhopadhyay, "Decision Making in Non-Stationary Environments with Policy-Augmented Search", Extended Abstract in the *23rd International Conference on Autonomous Agents and MultiAgent Systems (AAMAS 2024)*. (Acceptance rate: 40%) [code]

Shreyas Ramakrishna, **Baiting Luo**, Yogesh Barve, Gabor Karsai, Abhishek Dubey, "Risk-Aware Scene Sampling for Dynamic Assurance of Autonomous Systems", *IEEE International Conference on Assured Autonomy (ICAA 2022)* [code]

Xiangguo Liu, **Baiting Luo**, Ahmed Abdo, Nael Abu-Ghazaleh, Qi Zhu, "Securing Connected Vehicle Applications with An Efficient Dual Cyber-Physical Blockchain Framework", accepted by *32nd IEEE Intelligent Vehicles Symposium (IV 2021)*. (Acceptance rate: 49.3%)

Book Chapters

Dubey, A., Johnson, T. T., Koutsoukos, X., **Luo, B.**, Manzanas Lopez, D., Maroti, M., Mukhopadhyay, A., Potteiger, N., Serbinowska, S., Stojcsics, D., Zhang, Y., & Karsai, G. (in press). *Towards Assured Autonomy using Neuro-Symbolic Components and Systems*. In *Neuro-symbolic AI: Foundations and Applications* (Chap. 4). Wiley.

Awards and Honors

- Vanderbilt Provost's Pathbreaking Discovery Award, 2025
- Vanderbilt C.F. Chen Best Paper Runner-Up Award, 2024
- AAMAS Student Scholarship, 2024
- Vanderbilt University Graduate School Travel Grant, 2024
- Vanderbilt University Graduate School Travel Grant, 2023
- Northwestern Best MS Computer Engineering Thesis Award, 2021
- Dean's Graduate Fellowship, 2021
- Russell G. Hamilton Scholar, 2021

External Services

Conference Reviewer & Program Committee Member

- International Conference on Learning Representations (ICLR 2026/2027)
- The ACM Web Conference (WWW 2026)
- Neural Information Processing Systems (NeurIPS 2025/2026)
- Knowledge Discovery and Data Mining (KDD 2025/2026/2027)
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)
- International Joint Conference on Neural Networks (IJCNN 2025/2026)
- International Conference on Neural Information Processing (ICONIP 2024)
- IEEE International Transportation Systems Conference (ITSC 2022)
- Artifact Evaluation Committee, ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS 2024/2025/2026)
- Program Committee Member, International Conference on Data Mining and Big Data (DMBD 2024)

Journal Reviewer

- IEEE Transactions on Intelligent Vehicles
- ACM Transactions on Computing for Healthcare
- IEEE Internet of Things Journal

Core Coursework

Machine Learning, Advanced Machine Learning, Deep Learning, Advanced Deep Learning: Representation Learning, AI Programming, Reinforcement Learning, Design & Analysis of Algorithms, Data Structures, Computer Vision, Massively Parallel Programming w/ CUDA, Database Systems

Skills

Programming Languages: Python, C++, C, Java, SQL, LaTeX

Frameworks & Libraries: Scikit-Learn, PyTorch, TensorFlow, NumPy, Pandas, Mujoco